Physical Chemistry CUME – Kinetics
11/15/2024

1. **Rate Laws**: From the kinetic data below obtained for reaction $A + B \rightarrow C$, find the appropriate differential rate law for this reaction. Note: Rows 2 to 5 below represent $[A]_0$, $[B]_0$, $[C]_0$, and $[C]_{\Delta t}$, respectively ($\Delta t = 1\ s$). The data is shown for 4 different experiments. [40]

Experiment	1	2	3	4
$[A]_0$/ mol dm^{-3}	1.0000	1.0000	0.5000	0.5000
$[B]_0$/ mol dm^{-3}	1.0000	2.0000	1.0000	2.0000
$[C]_0$/ mol dm^{-3}	0.0000	0.5000	1.0000	2.0000
$[C]_{\Delta t}$/ mol dm^{-3}	0.0100	0.5071	1.0035	2.0025

2. **Complex Reactions**:
   a. Find a set of coupled differential equations describing the kinetics of reaction $A + B \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} AB \overset{k_2}{\rightarrow} C + D$. It is not necessary to simplify or combine the rate laws. [30]
   b. Using the steady state approximation for complex AB, find the rate of product formation in terms of the reactant concentrations. [30]